HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Collected and processed diverse datasets including macroeconomics, housing, and sector-specific information from web scraping and API interactions to support investment research.
- Developed R scripts to automate data wrangling and cleaning tasks, preparing approximately 50 GB of raw data weekly for analysis by senior researchers.
- Designed and implemented interactive data visualizations using **R (ggplot2, Tidyverse)** and **Power BI** to highlight key trends and insights for research reports.
- Assisted senior analysts in applying statistical techniques and predictive analytics, including time-series forecasting on market data, to identify emerging investment opportunities.
- Utilized **MS Excel** extensively for data manipulation, financial modelling, and generating ad-hoc reports for internal stakeholders and client presentations.
- Integrated generative AI tools, specifically **MS Copilot**, into the research workflow to accelerate content generation and explore novel data insights.
- Managed data storage and architecture, ensuring data integrity and accessibility for research projects across the team.

SKILLS

Programming Languages – R, Python, SQL, Java, MS Excel

Data Analysis & Visualization – Power BI, Tableau, ggplot2, Tidyverse, DAX, Power Query, Data Modelling, Statistical Analysis, Exploratory Data Analysis, Time-Series Analysis

Data Engineering & Web Scraping – Web Data Mining, API Interactions, JSON/XML Parsing, Data Wrangling, Data Cleaning, Data Warehousing Concepts

Machine Learning & AI – Predictive Analytics, Forecasting Techniques, Generative AI (MS Copilot), AI Application for Insights

Tools & Platforms – MS Office Suite (Word, PowerPoint), Git, Jenkins, AWS (EC2/S3/Lambda/CloudWatch), Docker, Linux

Finance Domain – Investment Banking Research Support, Macroeconomics, Sector Analysis, Company Analysis

PROJECTS

Investment Research Data Aggregation Platform

- Developed a Python-based system to aggregate financial data from multiple public APIs and web sources, parsing JSON and XML responses.
- Implemented data cleaning routines using pandas to handle missing values and standardize formats across disparate data feeds, processing over 10,000 data points daily.
- Built a PostgreSQL database schema to store and manage the collected financial and economic data, enabling efficient querying for research analysts.

Sector Performance Analysis Dashboard

- Created an interactive dashboard using **Power BI** to visualize historical performance data for key market sectors, incorporating DAX measures for trend analysis.
- Extracted and transformed data from relational databases and CSV files using Power Query, ensuring data accuracy for reporting purposes.
- Presented the finalized dashboard to the research team, facilitating quicker identification of sector-specific investment opportunities and risks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)